



GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Data Mining and Data Warehousing

DS-510-MJ

Interview Preparation — Topic-wise Q&A

Semester I · Elective

Curated interview questions and answers for Data Mining and Data Warehousing, organised by topic from the SPPU M.Sc. Data Science syllabus. Ideal for viva, placement and internship preparation.

Tip: read the answer, then close the PDF and try to explain each concept aloud in your own words.

Data warehousing & OLAP

Q. What is a data warehouse?

A central repository of integrated, subject-oriented, time-variant and non-volatile data used for decision support and analysis.

Q. What is OLAP?

Online Analytical Processing — enables multidimensional analysis with operations like roll-up, drill-down, slice and dice.

Q. Star vs snowflake schema?

A star schema has a central fact table linked to denormalised dimension tables; a snowflake schema normalises those dimensions into sub-tables.

Mining techniques

Q. Explain the Apriori algorithm.

It finds frequent itemsets using the principle that any subset of a frequent itemset must also be frequent, then generates association rules with support and confidence thresholds.

Q. What are support, confidence and lift?

Support = frequency of an itemset; confidence = $P(B|A)$ for rule $A \rightarrow B$; lift = confidence / $P(B)$, where lift > 1 indicates a positive association.

Q. Classification vs clustering?

Classification is supervised (predicts known labels); clustering is unsupervised (groups similar data without labels).

Q. How does a decision tree choose splits?

By maximising information gain (entropy) or minimising Gini impurity at each node.

